Does the triangular lattice minimise the harmonic zero-point energy? A numerical counterexample for completely monotone potentials, with kernel-checked curvature identities

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Abstract

For a pair potential $V(r) = g(r^2)$ and a planar Bravais lattice of covolume one, the harmonic zero-point energy is $E_{\mathrm{zp}} = \frac{1}{2}\langle \sum_j \omega_j(\mathbf{k})\rangle_{\mathrm{BZ}}$. Planar universal optimality (the Cohn–Kumar conjecture, a theorem for Gaussians by Montgomery; its general proof in openai/math is AI-generated and not peer reviewed) says that the triangular lattice minimises the *static* energy for every completely monotone g. The natural quantum analogue, conjecture **QU-1**, says it also minimises E_{zp} among unit-covolume planar Bravais lattices. We tested it before trying to prove it, and it fails numerically (Tier X, below): we do not claim a proof that it is false.

Tier X (floating point). For the Gaussian $g(t) = e^{-\pi t}$, the rhombic lattice $\tau = 0.5 + 0.966025\,i$ is dynamically stable and has $\Delta E_{\mathrm{zp}} = -1.31 \times 10^{-2}$ relative to the triangular lattice ($E_{\mathrm{zp}} = 1.4902783183$ against 1.5033523522). Two formulas in four implementations agree: a real-space folded-grid Rust code and an independent numpy re-implementation of it; a reciprocal-space (Poisson) Rust code with a spectrally convergent zone rule (quadrature error about 4.4×10^{-15}); and a short third Poisson implementation in numpy. The conjecture is numerically refuted for 8 of 12 completely monotone potentials tested (Gaussian, Yukawa, dipolar-exciton, Riesz). One of the eight, `riesz:4`, is weak: it rests on a single competitor whose exact long-wave margin is 0.08 % of the triangular value, its finite-$\mathbf{k}$ stability comes from the G1 grids only, and its triangular point is a local minimum ($c > 0$). Four potentials (Gaussian $a = 0.25, 2, 4$ and Riesz $s = 6$) are not refuted, on a sample and two dense paths, which is not an exhaustive search. At the triangular point the curvature of E_{zp} is isotropic and, for the Gaussian $e^{-\pi a r^2}$, numerically negative on $a \in (0.37668, 1.58148)$ among $0.15 \le a \le 4$ (a scan of 23 values of $a \in [0.1, 8]$ followed by bisection of the sign changes; gaps between scan points are not excluded).

Tier A (Lean 4.34.1, Mathlib; axioms within {`propext`, `Classical.choice`, `Quot.sound`}; planted `sorry` and planted-axiom controls caught). 19 finite and algebraic statements: derivatives and Laplacian identities of the lattice-vector length Q, the Gaussian curvature densities, the 120-degree rotation lemmas that force an isotropic Hessian at $\rho = e^{i\pi/3}$, and an inversion step for the mean-square frequency sum rule. They certify none of the statements about E_{zp}.

Tiers L and C. Cited: Montgomery's theorem (via Deng–Luo) and two results of Luo–Wei, with the Elenchus citation gate accepting the three quotations. Human arguments: the sum rule $\langle \mathrm{tr}\, D\rangle = 4\pi a - (4\pi^2/a)S(1/a)$, the Einstein bound, and a concavity decomposition of the curvature. No result here rests on openai/math, apart from the pinned toolchain. We claim no proof of anything about E_{zp} and no breakthrough; the contribution is a negative result with controls, and a restricted conjecture QU-1′ (Tier X-supported) for a range of Gaussians.

Tiers. [A]: kernel-checked Lean, axioms within the allowlist, planted controls caught. [L]: a cited source, quoted in the ledger and run through the Elenchus citation gate. [C]: an argument or human proof that is not machine-checked. [X]: floating-point exploration, which never supports a claim by itself. Identifiers such as `ZP-X-0001` refer to `ledger/ledger.json` (Section 6). A Tier A tag attaches to the Lean statement, never to its physical gloss. Statements about E_{zp} are Tier X or C throughout.

1 Setting, the conjecture, and what was preregistered

The functional. Let L_τ be the planar lattice of shape $\tau = x + iy$ ($y > 0$) with basis $a_1 = (1/\sqrt{y}, 0)$, $a_2 = (x/\sqrt{y}, \sqrt{y})$, covolume one. Particles of unit mass sit on L_τ and interact by $V(r) = g(r^2)$ (units $\hbar = m = 1$). In the harmonic approximation

$$D(\mathbf{k}) = \sum_{R \in L \setminus 0} (1 - \cos \mathbf{k}\cdot R) \operatorname{Hess} V(R), \qquad \omega_j(\mathbf{k}) = \sqrt{\lambda_j(D(\mathbf{k}))}, \qquad E_{\mathrm{zp}}(\tau) = \tfrac{1}{2}\Big\langle \sum_{j=1}^{2} \omega_j \Big\rangle_{\mathrm{BZ}} \quad [\mathsf{C},\ \mathsf{ZP\text{-}C\text{-}0001}].$$

A lattice is *dynamically stable* if $D(\mathbf{k}) \succeq 0$ for all $\mathbf{k}$. E_{zp} is invariant under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$ (the lattice is unchanged up to rotation), which we tested numerically.

Conjecture QU-1 (zero-point version of triangular optimality). For every completely monotone g (a positive mixture of e^{-ut} in $t = r^2$), the triangular lattice $\rho = e^{i\pi/3}$ minimises $E_{\mathrm{zp}}(\tau)$ over unit-covolume planar Bravais lattices. It was proposed in the programme document of 2026-10-25 as a quantum companion of planar universal optimality (the static statement, Cohn–Kumar; its proof by the openai/math release is AI-generated and not peer reviewed, and nothing here depends on it). The static statement for the Gaussian is Montgomery's theorem [1, 2] [L, ZP-L-0003]. QU-1 was preregistered with a prior of 0.6 that it survives a falsification gate (G1), to be followed by a proof attempt for Gaussians (G2) *only if* G1 survived.

What happened, and deviations from the preregistration. G1 numerically refuted QU-1 (Tier X). G2 as preregistered was therefore not run. What was run under the label G2 (an independent re-verification of the counterexample, the curvature scan, the mechanism analysis and the Lean module) is **post hoc**: the re-verification is a robustness check, while the scan of $c(a)$, its sign changes, the decomposition and QU-1′ are exploratory and were not preregistered. Further deviations: (i) *unstable-lattice policy.* A lattice with any eigenvalue of D below -10^{-9} times the scale is dynamically unstable and its E_{zp} is set to NaN and excluded; no clipping of imaginary frequencies is used for any verdict (a competitor-favouring clipped value is reported only in the data files). The policy is a scope decision made at G1. (ii) The preregistration asked for a neutralising background for long-range potentials; Coulomb (Riesz $s = 1$) and Keldysh need a plasmon treatment of $k \to 0$ that the code does not have, so they are *excluded* and nothing is claimed for them. The dipolar-exciton and Riesz $s > 2$ potentials are untruncated with $1/r^3$ or faster tails, for which no background is needed. (iii) The Lean module and this paper were not preregistered.

2 Step-by-step human demonstration, with the matching Lean statements

The refutation itself is a numerical inequality, so this section cannot prove it; it proves, step by step, everything that is exact and used by the analysis, and says where each step is machine-checked. Listings show Lean statements verbatim; proofs are elided unless marked (full source and hash in Section 3). Throughout, $Q_{mn}(\tau) = |ma_1 + na_2|^2 = ((m + nx)^2 + n^2y^2)/y$ and $\Delta = y^2(\partial_x^2 + \partial_y^2)$ is the hyperbolic Laplacian.

Step 0: the model [C, ZP-C-0001]. See Section 1. For $V = e^{-\pi a r^2}$, $\operatorname{Hess} V(R) = (-2\pi a\, I + 4\pi^2 a^2 R R^{\mathsf{T}}) e^{-\pi a |R|^2}$.

Step 1: the length function [A, ZP-A-0001]. $\partial_x Q = 2n(m+nx)/y$, $\partial_y Q = n^2 - (m+nx)^2/y^2$, $\partial_x^2 Q = 2n^2/y$, $\partial_y^2 Q = 2(m+nx)^2/y^3$. *Proof:* differentiate the quotient N/y with $N = (m+nx)^2 + n^2y^2$ in each variable.

```
noncomputable def Q (m n x y : ℝ) : ℝ := ((m + n * x) ^ 2 + n ^ 2 * y ^ 2) / y

theorem hasDerivAt_Q_x (m n x y : ℝ) :
    HasDerivAt (fun t => Q m n t y) (2 * n * (m + n * x) / y) x := ... -- proof elided

theorem hasDerivAt_Q_y (m n x y : ℝ) (hy : y ≠ 0) :
    HasDerivAt (fun t => Q m n x t) (n ^ 2 - (m + n * x) ^ 2 / y ^ 2) y := ... -- proof elided

theorem hasDerivAt_Qx_x (m n x y : ℝ) :
    HasDerivAt (fun t => 2 * n * (m + n * t) / y) (2 * n ^ 2 / y) x := ... -- proof elided

theorem hasDerivAt_Qy_y (m n x y : ℝ) (hy : y ≠ 0) :
    HasDerivAt (fun t => n ^ 2 - (m + n * x) ^ 2 / t ^ 2) (2 * (m + n * x) ^ 2 / y ^ 3) y := ... -- proof
     elided
```

Step 2: two identities [A, ZP-A-0002]. $y^2(Q_{xx} + Q_{yy}) = 2Q$ and $y^2(Q_x^2 + Q_y^2) = Q^2$. *Proof:* insert Step 1; with $u = m + nx$, $y^2(2n^2/y + 2u^2/y^3) = 2(u^2 + n^2y^2)/y = 2Q$ and $y^2\big(4n^2u^2/y^2 + (n^2 - u^2/y^2)^2\big) = (u^2 + n^2y^2)^2/y^2 = Q^2$.

```
theorem lap_Q (m n x y : ℝ) (hy : y ≠ 0) :
    y ^ 2 * (2 * n ^ 2 / y + 2 * (m + n * x) ^ 2 / y ^ 3) = 2 * Q m n x y := ... -- proof elided

theorem grad_Q (m n x y : ℝ) (hy : y ≠ 0) :
    y ^ 2 * ((2 * n * (m + n * x) / y) ^ 2 + (n ^ 2 - (m + n * x) ^ 2 / y ^ 2) ^ 2) = (Q m n x y) ^ 2 := ...
     -- proof elided
```

Step 3: curvature of $f(Q)$ [A for the algebra, C for the chain rule; ZP-A-0003, ZP-C-0003]. By the chain rule $\Delta f(Q) = y^2[f''(Q)(Q_x^2 + Q_y^2) + f'(Q)(Q_{xx} + Q_{yy})] = Q^2 f''(Q) + 2Q f'(Q) = h(Q)$, with $h(u) = (u^2 f'(u))'$. The second equality is Step 2 and is the Lean statement below with $f_1 = f'(Q)$, $f_2 = f''(Q)$. The chain rule and termwise differentiation of the absolutely convergent sum are *not* formalised, so for a pair energy $F(\tau) = \sum_{R \neq 0} f(|R|^2)$ the formula $\Delta F = \sum_{R \neq 0} h(|R|^2)$ is [C].

```
theorem lap_comp (m n x y f1 f2 : ℝ) (hy : y ≠ 0) :
    y ^ 2 * (f2 * ((2 * n * (m + n * x) / y) ^ 2 + (n ^ 2 - (m + n * x) ^ 2 / y ^ 2) ^ 2)
        + f1 * (2 * n ^ 2 / y + 2 * (m + n * x) ^ 2 / y ^ 3))
      = (Q m n x y) ^ 2 * f2 + 2 * Q m n x y * f1 := ... -- proof elided
```

Step 4: the Gaussian densities [A, ZP-A-0004]. For $f(u) = e^{-\pi a u}$: $f' = -\pi a\, e^{-\pi a u}$ and $h(u) = (-2\pi a u + \pi^2 a^2 u^2)e^{-\pi a u}$. For the second-moment density $m_2(u) = \operatorname{tr}\operatorname{Hess} V = 4(ug')' = (-4\pi a + 4\pi^2 a^2 u)e^{-\pi a u}$ one gets $h(u) = (16\pi^2 a^2 u - 20\pi^3 a^3 u^2 + 4\pi^4 a^4 u^3)e^{-\pi a u}$. *Proof:* product rule, then collect terms.

```
theorem gauss_f' (a u : ℝ) :
    HasDerivAt (fun v => Real.exp (-(Real.pi * a) * v)) (-(Real.pi * a) * Real.exp (-(Real.pi * a) * u)) u :=
     ... -- proof elided

theorem gauss_h (a u : ℝ) :
    HasDerivAt (fun v => v ^ 2 * (-(Real.pi * a) * Real.exp (-(Real.pi * a) * v)))
      ((-2 * Real.pi * a * u + Real.pi ^ 2 * a ^ 2 * u ^ 2) * Real.exp (-(Real.pi * a) * u)) u := ... -- proof
      elided

noncomputable def m2f (a u : ℝ) : ℝ := (-4 * Real.pi * a + 4 * Real.pi ^ 2 * a ^ 2 * u) * Real.exp (-(Real.pi
     * a) * u)

noncomputable def m2f' (a u : ℝ) : ℝ :=
  (8 * Real.pi ^ 2 * a ^ 2 - 4 * Real.pi ^ 3 * a ^ 3 * u) * Real.exp (-(Real.pi * a) * u)

theorem m2_f' (a u : ℝ) : HasDerivAt (fun v => m2f a v) (m2f' a u) u := ... -- proof elided

theorem m2_h (a u : ℝ) :
    HasDerivAt (fun v => v ^ 2 * m2f' a v)
      ((16 * Real.pi ^ 2 * a ^ 2 * u - 20 * Real.pi ^ 3 * a ^ 3 * u ^ 2 + 4 * Real.pi ^ 4 * a ^ 4 * u ^ 3)
```

```
        * Real.exp (-(Real.pi * a) * u)) u := ... -- proof elided
```

Step 5: the triangular point and its stabiliser [A, ZP-A-0005]. Put $\rho = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. Then $\rho^2 = \rho - 1$, so $\gamma(\tau) = (\tau - 1)/\tau$, the action of $\left(\begin{smallmatrix} 1 & -1 \\ 1 & 0 \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$, fixes ρ, and $\gamma'(\rho) = \rho^{-2} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, the rotation by $-120°$. *Proof:* $(\rho - 1)/\rho = \rho^2/\rho = \rho$; and $\rho^2 \cdot (-\frac{1}{2} - i\frac{\sqrt{3}}{2}) = (\rho - 1)(-\frac{1}{2} - i\frac{\sqrt{3}}{2}) = (-\frac{1}{2} + i\frac{\sqrt{3}}{2})(-\frac{1}{2} - i\frac{\sqrt{3}}{2}) = \frac{1}{4} + \frac{3}{4} = 1$, using $(\sqrt{3}/2)^2 = 3/4$. The derivative of $(\tau - 1)/\tau = 1 - 1/\tau$ is $1/\tau^2$.

```
noncomputable def rho : ℂ := ⟨1 / 2, Real.sqrt 3 / 2⟩

theorem rho_sq : rho ^ 2 = rho - 1 := ... -- proof elided

theorem sqrt3_half_sq : (Real.sqrt 3 / 2) ^ 2 = 3 / 4 := ... -- proof elided

theorem gamma_fixes_rho : (rho - 1) / rho = rho := by
  rw [← rho_sq, sq, mul_div_assoc, div_self rho_ne_zero, mul_one]

theorem rho_sq_mul_rot : rho ^ 2 * (⟨-1 / 2, -(Real.sqrt 3 / 2)⟩ : ℂ) = 1 := ... -- proof elided

theorem hasDerivAt_gamma : HasDerivAt (fun τ : ℂ => (τ - 1) / τ) (1 / rho ^ 2) rho := ... -- proof elided
```

Step 6: isotropy at ρ [A for the linear algebra, C for the glue; ZP-A-0006, ZP-C-0002]. A vector fixed by the $120°$ rotation is zero, and a symmetric 2×2 matrix commuting with it is scalar. *Proof:* with $s^2 = 3/4$ the two entries of $RH = HR$ give $sq = 0$ and $s(r - p) = 0$, and $s \neq 0$. *Glue (not formalised):* $E_{\mathrm{zp}}(\gamma\tau) = E_{\mathrm{zp}}(\tau)$ and $\gamma\rho = \rho$ give, for $E_{\mathrm{zp}} \in C^2$ near ρ, $J^{\mathsf{T}}\nabla E_{\mathrm{zp}} = \nabla E_{\mathrm{zp}}$ and $J^{\mathsf{T}}HJ = H$ with J the rotation. Hence $\nabla E_{\mathrm{zp}}(\rho) = 0$ and $H = \kappa I$, so

$$E_{\mathrm{zp}}(\tau) - E_{\mathrm{zp}}(\rho) = c\,\frac{|\tau - \rho|^2}{y_\rho^2} + O(|\tau - \rho|^3), \qquad c = \tfrac{1}{4}\Delta E_{\mathrm{zp}}(\rho).$$

The sign of c decides local minimum ($c > 0$) or local maximum ($c < 0$). The same argument applies to the static energy and to any other $SL_2(\mathbb{Z})$-invariant functional.

```
theorem rot120_comm_symm (p q r s : ℝ) (hs : s ^ 2 = 3 / 4)
    (h11 : -1 / 2 * p + -s * q = p * (-1 / 2) + q * s)
    (h12 : -1 / 2 * q + -s * r = p * -s + q * (-1 / 2)) :
    q = 0 ∧ r = p := ... -- proof elided

theorem rot120_fixed_vec (g1 g2 s : ℝ) (hs : s ^ 2 = 3 / 4)
    (e1 : -1 / 2 * g1 + -s * g2 = g1) (e2 : s * g1 + -1 / 2 * g2 = g2) :
    g1 = 0 ∧ g2 = 0 := by
  have h1 : g1 = 0 := by linear_combination (s * e2 - 3 / 2 * e1 - g1 * hs) / 3
  refine ⟨h1, ?_⟩
  linear_combination (-(s * e1) - 3 / 2 * e2 - g2 * hs) / 3
```

Step 7: the mean-square frequency [C with A and L; ZP-C-0005, -0006, -0007]. Averaging $\operatorname{tr} D$ over the zone kills every $\cos \mathbf{k}\cdot R$ with $R \neq 0$: $\langle \operatorname{tr} D \rangle_{\mathrm{BZ}} = \sum_{R \neq 0} \Delta V(R)$. Poisson summation (the dual of a unit-covolume planar lattice is a rotation of it) turns this into

$$\langle \operatorname{tr} D \rangle = 4\pi a - \frac{4\pi^2}{a}\, S(1/a; L), \qquad S(\alpha; L) = \sum_{P \in L} |P|^2 e^{-\pi\alpha|P|^2}.$$

(i) Luo and Wei prove that for $\alpha \geq 1$ the minimum of $S(\alpha; \cdot)$ over unit-density lattices is attained at the hexagonal lattice [3] [L, ZP-L-0001]. Hence, for $a \leq 1$, the triangular lattice is a global *maximiser* of $\langle \operatorname{tr} D \rangle$ [C, ZP-C-0007]. *(ii)* At $a = 1$, $\theta_L(\alpha) = \sum_P e^{-\pi\alpha|P|^2}$ satisfies $\theta_L(1/\alpha) = \alpha\theta_L(\alpha)$ (Jacobi; classical, not formalised), so the Lean step below gives $\theta_L'(1) = -\theta_L(1)/2$. Since $S(1) = -\theta_L'(1)/\pi$, we get $S(1) = \theta_L(1)/2\pi$ (this is also Luo–Wei Lemma 3.5 [L, ZP-L-0002]) and $\langle \operatorname{tr} D \rangle = 4\pi - 2\pi\theta_L(1)$; with Montgomery's theorem the triangular lattice is the unique maximiser at $a = 1$ [C, ZP-C-0006].

```
theorem deriv_at_one_of_inversion (f : ℝ → ℝ) (d : ℝ) (hf : HasDerivAt f d 1)
    (hinv : ∀ α : ℝ, 0 < α → f (1 / α) = α * f α) : d = -f 1 / 2 := ... -- proof elided
```

Step 8: two inequalities [C, ZP-C-0008, ZP-C-0009]. *Einstein bound.* By Cauchy–Schwarz, $\frac{1}{2}(\omega_1+\omega_2) \le \sqrt{(\omega_1^2+\omega_2^2)/2}$ pointwise, and Jensen's inequality over the zone gives $E_{\rm zp} \le \sqrt{\langle \operatorname{tr} D\rangle/2}$, with equality only for a flat spectrum. By Step 7, the right side is largest at the triangular lattice for $a \le 1$. This makes QU-1 (a *minimum* at ρ) look less natural for wide Gaussians, but an upper bound that is largest at ρ does not locate the extrema of $E_{\rm zp}$ itself. *Concavity.* Write $\Phi(D) = \operatorname{tr} D^{1/2}$ at fixed reduced $\mathbf{k}$. Then $\Delta E_{\rm zp} = \mathrm{Lin} + \mathrm{Conc}$, where $\mathrm{Lin} = \frac{1}{2}\langle y^2 \sum_i \operatorname{tr}(\Phi'(D)\partial_i^2 D)\rangle$ is a linear-response term and $\mathrm{Conc} = \frac{1}{2}\langle y^2 \sum_i \Phi''(D)[\partial_i D, \partial_i D]\rangle \le 0$ by operator concavity of the square root. Nothing in Steps 1–7 controls the sign of Lin, and the sum rule alone does not decide the sign of c (Table 5). Because $E_{\rm zp}$ is a trace of a square root of a lattice sum and not a pair sum, the Bernstein-mixture argument that carries Montgomery's Gaussian theorem to every completely monotone potential is linear in V and does not apply [C, ZP-C-0010].

Step 9: what a human cannot yet prove, and how to check it [X, ZP-X-0001]. The statement "$E_{\rm zp}(0.5 + 0.966025\,i) < E_{\rm zp}(\rho)$ for $a = 1$ and the first lattice is dynamically stable" has no hand proof here and no Lean statement: it needs the zone integral of $\sqrt{\lambda}$. A reader can check it with the listing below (Poisson form of Step 7, a half-shifted $n \times n$ grid, Richardson in n^{-3}, which is the error of the conical singularity at Γ), which shares no code with the other implementations.

```
b1, b2 = (sqrt(y), -x/sqrt(y)), (0, 1/sqrt(y))                # dual basis, covolume 1
kap = (i+.5)/n * b1 + (j+.5)/n * b2                           # k = 2 pi kap, i,j = 0..n-1
q   = xi + kap  for xi = m*b1 + p*b2, |m|,|p| <= 7            # dual lattice points
D   = (4 pi^2/a) * ( sum_xi  q q^T exp(-pi |q|^2/a)  -  sum_xi  xi xi^T exp(-pi |xi|^2/a) )
E   = mean( 0.5 * sqrt( tr D + 2 sqrt(det D) ) )              # = (1/2)(sqrt(l1) + sqrt(l2))
E_zp = (8 E(n=256) - E(n=128)) / 7                            # human_check.py has the full code
```

At $a = 1$ it gives $E_{\rm zp} = 1.490278318$ for $\tau = 0.5 + 0.966025\,i$ and 1.503352352 for ρ, so $\Delta E_{\rm zp} = -1.3074 \times 10^{-2}$, which agrees with the zone rule of Section 4 to 4.3×10^{-12}. On the sampled grids, $\min \lambda/|\mathbf{k}|^2$ is 0.068 (rhombic) and 0.097 (triangular), positive; the square lattice gives -0.1441 (unstable, the negative control). *Convention.* The full code (`human_check.py`; the listing omits this line) divides by $|\mathbf{k}|^2$ of the nearest equivalent dual-lattice point, i.e. $\mathbf{k}$ reduced to the Wigner–Seitz cell, whereas Table 2 and the Rust code use the parallelogram $[-\frac{1}{2}, \frac{1}{2}]^2$ (0.03493 rhombic, 0.03222 triangular, minimised over the zone). The two conventions give different numbers, and even a different order of the two lattices, because $\lambda/|\mathbf{k}|^2$ depends on the reduction of $\mathbf{k}$; only the sign is used, and it is the same. A grid check of stability is necessary, not sufficient; the zone minimum and the exact long-wave limit are in Table 2.

3 Lean formalization

Module. `formal_cert/Programmes/ZeroPoint/Basic.lean` (namespace `ProgZeroPoint`; 19 theorems printed; `grep -c sorry` gives 0; it imports Mathlib only), with planted controls in a separate file `Controls.lean`. Both compile with the pinned stack (Lean 4.34.1, Mathlib `d13f23b7`); the pinned openai/math checkout supplies the toolchain and nothing else. The lane compiled `Basic.lean` with `rc` 0 in 8.5 s. All 19 printed declarations have axioms within the allowlist (19 of 19), and no `sorryAx`. (The helper `rho_ne_zero` is used by `gamma_fixes_rho`; it is not printed separately, and its axioms are included in those of `gamma_fixes_rho`.) Verbatim output of `#print axioms`:

```
rc = 0, 8.5 s
'ProgZeroPoint.deriv_at_one_of_inversion' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.hasDerivAt_Q_x' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.hasDerivAt_Q_y' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.hasDerivAt_Qx_x' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.hasDerivAt_Qy_y' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.lap_Q' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.grad_Q' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.lap_comp' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.gauss_f'' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.gauss_h' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.m2_f'' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.m2_h' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.rot120_comm_symm' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.rot120_fixed_vec' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.sqrt3_half_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.rho_sq' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.gamma_fixes_rho' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.rho_sq_mul_rot' depends on axioms: [propext, Classical.choice, Quot.sound]
'ProgZeroPoint.hasDerivAt_gamma' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Planted controls [A, ZP-A-0008]. The file below "proves" the false identity $y^2(Q_{xx} + Q_{yy}) = Q$ (factor 2 dropped) with a `sorry` and with a planted axiom. It compiles with `rc` 0 *despite* both, so the exit code is not the

gate; the axiom list is. `ctl_sorry` depends on `sorryAx` (yes) and `ctl_axiom` on the planted `ctl_cheat` (yes); `lap_wrong_false` is an honest theorem refuting the planted statement (instance $m = 1, n = 0, x = 0, y = 1$, $Q = 1$, left side 2), within the allowlist (yes).

```
theorem ctl_sorry : ∀ m n x y : ℝ, y ≠ 0 →
    y ^ 2 * (2 * n ^ 2 / y + 2 * (m + n * x) ^ 2 / y ^ 3) = Q m n x y := sorry

axiom ctl_cheat : False

theorem ctl_axiom : ∀ m n x y : ℝ, y ≠ 0 →
    y ^ 2 * (2 * n ^ 2 / y + 2 * (m + n * x) ^ 2 / y ^ 3) = Q m n x y := ctl_cheat.elim

theorem lap_wrong_false : ¬ (∀ m n x y : ℝ, y ≠ 0 →
    y ^ 2 * (2 * n ^ 2 / y + 2 * (m + n * x) ^ 2 / y ^ 3) = Q m n x y) := by
  intro h
  have := h 1 0 0 1 one_ne_zero
  norm_num [Q] at this
```

```
rc = 0, 5.7 s
formal_cert/Programmes/ZeroPoint/Controls.lean:14:8: warning: declaration uses `sorry`
'ProgZeroPoint.ctl_sorry' depends on axioms: [propext, sorryAx, Classical.choice, Quot.sound]
'ProgZeroPoint.ctl_axiom' depends on axioms: [propext, Classical.choice, ProgZeroPoint.ctl_cheat, Quot.sound]
'ProgZeroPoint.lap_wrong_false' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Provenance and status. sha256 of the two files (both match the values recorded when the lane compiled them):

- `Basic.lean`: `7d59f1817767ae64cc37d017348e1eb5f5b9ed83ed068086fae839baff0d64f0`
- `Controls.lean`: `c725bdcba93362681c58f65eaf00d717eea37254c10f364bab4e039d645acd90`

At paper stage the same pipeline recompiled both files (`Basic.lean` rc 0 in 7.9 s, with the `-olean` build; `Controls.lean` rc 0); the axiom output is identical to the lane's for `Basic.lean` and identical to it for `Controls.lean`. This is a rerun by the same AI pipeline, not a separate verifier; the separate AI audit of Section 8 recompiled both files independently and got the same axiom lists. The Elenchus seed detector (`tools/elenchus_check.py`) *could not run* on these files: it skipped both with exit code 2, "toolchain mismatch: lean-toolchain pins 4.34.0-rc2, lean reports 4.34.1" (its own toolchain has no Mathlib). The axiom audit therefore rests on `#print axioms` through the pinned runner.

What Tier A certifies. Only the statements above: derivatives of Q and of the Gaussian densities, two Laplacian identities, the algebra of `lap_comp` (not the chain rule), 2×2 rotation algebra, Möbius facts at ρ, and the inversion step. *It does not certify any statement about E_{zp}, about lattice sums, about the counterexample, or about the physical gloss.*

4 Numerics and the rusty-SUNDIALS improvement

Everything in this section is [X]. Code: crate `qf-yukawa-crystal` of rusty-SUNDIALS, branch `physics/p2-zero-point-optimality-2026-10-25` (not pushed), commits `a961a71` (lattice-phonon module and falsification gate), `5f2f73b` (stability-edge verification, dense scans) and `6ad3cf7` (spectral zone rule); full hashes in Section 9.

Method 1 (G1, real-space). $D(\mathbf{k}) = \sum_{R \neq 0} (1 - \cos \mathbf{k} \cdot R) \operatorname{Hess} V(R)$ summed over $|R| < r_c = 400$ with a continuum tail, on Gamma-centred $n \times n$ grids ($n = 96, 192, 384$); the folding $\mathbf{k} \cdot R = 2\pi(im + jl)/n$ makes D for all $\mathbf{k}$ a separable transform, $O(n^3)$. The integrand has a conical singularity at Γ, so the grid error is $\propto n^{-3}$ (fitted exponent 3.00–3.01; error at $n = 96$ between 2.6×10^{-8} and 6.5×10^{-6}; 1.3×10^{-6} for `gauss:1`). Shapes: 167 coarse and 400 random with $0 \leq \operatorname{Re} \tau \leq \frac{1}{2}$, $\operatorname{Im} \tau \in [0.85, 2.4]$ (567 per potential), plus patches around ρ and i, then dense 0.005-step scans along $|\tau| = 1$ and $\operatorname{Re} \tau = \frac{1}{2}$. A competitor counts only if stable on $n = 96, 192, 384$ (and, for the two candidate points, $r_c = 400, 800$).

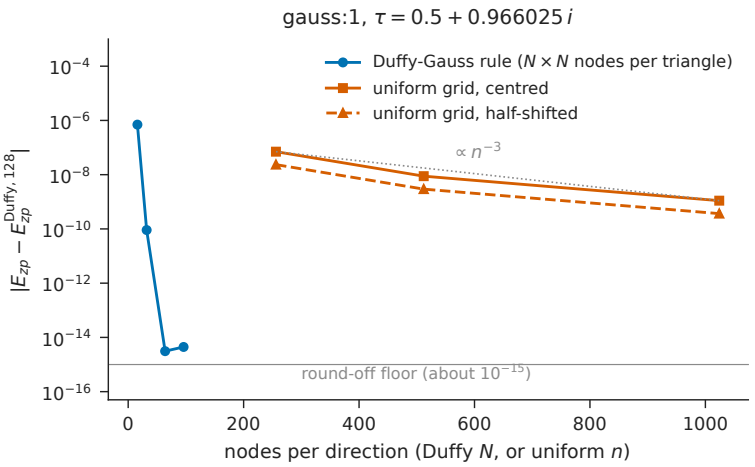


Figure 1: Error of the zone integral for `gauss:1` at $\tau = 0.5 + 0.966025\,i$ against the Duffy $N = 128$ value: spectral convergence of the new rule against the n^{-3} grid error (centred and half-shifted grids); n^{-3} line for reference [X]. Generated by `make_figures.py` from `data/poisson_check.json`.

Method 2 (G2, reciprocal space; the new rusty-SUNDIALS module). For the Gaussian, Poisson summation gives (Tier C, ZP-C-0004)

$$D(\mathbf{k}) = \frac{4\pi^2}{a} \sum_{\xi \in L^*} \Big[(\xi+\kappa)(\xi+\kappa)^{\mathsf{T}} e^{-\pi|\xi+\kappa|^2/a} - \xi\xi^{\mathsf{T}} e^{-\pi|\xi|^2/a} \Big], \quad \kappa = \mathbf{k}/2\pi.$$

The zone $[-\frac{1}{2}, \frac{1}{2}]^2$ is split into four triangles with apex at Γ; a Duffy substitution (radial coordinate s, Jacobian $s/2$) cancels the conical singularity $\sqrt{D} \sim |\mathbf{k}|$, and the integrand $\sqrt{\lambda_1} + \sqrt{\lambda_2} = \sqrt{\operatorname{tr} D + 2\sqrt{\det D}}$ is smooth across branch crossings, so Gauss–Legendre converges spectrally (Figure 1). Stability is tested by minimising $\lambda_{\min}(D)/|\mathbf{k}|^2$ over the zone (seed grid plus Nelder–Mead, $|\mathbf{k}| \geq 3 \times 10^{-3}$) and by the exact long-wave elastic tensor $C(t) = \frac{1}{2}\sum_R (t\cdot R)^2 \operatorname{Hess} V(R)$ over 3600 directions with no $\mathbf{k}$-grid. A first run without the lower bound on $|\mathbf{k}|$ flagged the marginal point of Table 2 unstable (ratio -3.24×10^{-5} at $|\mathbf{k}| \approx 5 \times 10^{-7}$); this was round-off ($\lambda \sim 10^{-17}$, below machine precision of a difference of $O(1)$ sums). Both result files are kept.

Controls. *Positive.* (a) Frozen phonon: the second difference of an independent pair-sum energy of a 5×5 supercell (values of V only; for the dipolar potential the Ewald supercell energy) equals $\frac{1}{2} e^{\mathsf{T}} D e$, max relative difference 2.14×10^{-7} over 33 cases. (b) Riesz statics against Epstein closed forms agree to 3.89×10^{-7} (and 9.7×10^{-10} for $s \geq 4$); dipolar statics against the Ewald sum to 3.89×10^{-7}. (c) $r_c = 400 \to 800$ changes E_{zp} by $\leq 2.2 \times 10^{-13}$; E_{zp} at $\tau, \tau+1, -1/\tau$ agrees to $\leq 2.2 \times 10^{-13}$. (d) Poisson D against the explicit real-space sum at 200 random $\mathbf{k}$ in 12 cases: $\leq 1.4 \times 10^{-14}$ relative; the exact sum rule against the real-space sum $\leq 5.9 \times 10^{-15}$ and against the Duffy $\langle \operatorname{tr} D \rangle \leq 2.7 \times 10^{-14}$. (e) An independent numpy implementation of the real-space method (explicit sum, no folding, centred and shifted grids) gives 1.5033524 at ρ for `gauss:1`, against 1.5033523522. (f) The static minimiser over the 567 shapes is the triangular lattice for all 12 potentials, as Montgomery's theorem and its Bernstein-mixture extensions predict; a violation would have signalled a code bug. (g) Stencil Laplacian of E_{static} and of $\langle \operatorname{tr} D \rangle$ against the exact lattice sum of h (Step 4): $\leq 2.5 \times 10^{-10}$ relative for $a \geq 0.5$ (ZP-X-0006). *Negative.* (h) Planted bug, dropping the $-\xi\xi^{\mathsf{T}}$ term: the real-space match fails by a relative 0.027 to 17.7; planted $a \leftrightarrow 1/a$ swap: invisible at the self-dual $a = 1$ (1.1×10^{-15}) but 0.904 to 9.4 elsewhere. (i) The square lattice is flagged unstable for every potential ($\min \lambda/|\mathbf{k}|^2 = -0.1475$ for `gauss:1`). (j) `gauss:2`, which survives G1, has no stable lower lattice at the `gauss:1` competitor shapes ($\Delta E_{\mathrm{zp}} = +4.533 \times 10^{-3}$ at $0.5 + 0.966\,i$; $0.295 + 0.9555\,i$ and $0.3 + 0.9539\,i$ unstable). (k) *Non-completely-monotone potentials.* For attractive Gaussian wells at the square-lattice shell distances (`wells:1.0;1.414;0.12;0.5`) the square lattice is a stable static minimiser lower than the triangular one by -0.506, and a stable rhombic lattice has E_{zp} lower by -3.03; a second member has the triangular static minimum but a stable lattice -2.04 lower in E_{zp}

Table 1: Non-completely-monotone controls (G1, $n = 96$, 567 shapes). "stable" counts dynamically stable shapes; ΔE_{zp} is the lowest stable competitor against the triangular lattice [X, ZP-X-0009].

potential	tri. stable	stable shapes	min $\Delta E_{\mathrm{static}}$	min ΔE_{zp} (stable)
`gdiff:0.3;1.5;0.5`	no	0/567	-2.40×10^{-2} at $(0.021, 2.137)$	none stable
`lj:0.9`	yes	5/567	0 at $(0.500, 0.866)$	$+1.26 \times 10^{-1}$ at $(0.450, 0.893)$
`morse:3;1.0`	yes	5/567	0 at $(0.500, 0.866)$	-2.04×10^{-2} at $(0.500, 0.966)$
`wells:1.0;1.414;0.12;0.5`	yes	28/567	-5.06×10^{-1} at $(0.000, 1.000)$	-3.03×10^{0} at $(0.150, 1.189)$
`wells:1.0;1.414;0.2;0.0`	yes	22/567	0 at $(0.500, 0.866)$	-2.04×10^{0} at $(0.441, 1.119)$

Table 2: `gauss:1`, $V = e^{-\pi r^2}$ [X, ZP-X-0001, ZP-X-0004]. E_{zp} from the Duffy rule ($N = 128$, error about 4.4×10^{-15}); $\min \lambda/|\mathbf{k}|^2$ over the zone ($|\mathbf{k}| \geq 3 \times 10^{-3}$); exact long-wave eigenvalue of the elastic tensor. †: marginally stable; not the headline.

lattice τ	E_{zp}	ΔE_{zp}	$\min \lambda/|\mathbf{k}|^2$	long-wave
triangular ρ	1.5033523522	0	0.03222	0.23826
$0.5 + 0.966025\,i$ (headline)	1.4902783183	-1.3074×10^{-2}	0.03493	0.10466
$0.3 + 0.953939\,i$	1.4546849815	-4.8667×10^{-2}	0.00498	0.00498
$0.295 + 0.9555\,i$ (†)	1.4521219619	-5.1230×10^{-2}	0.00013	0.00013
i (square, negative control)	–	–	-0.1475 (unstable)	-0.1475

(Table 1); `morse:3;1.0` also has a stable competitor ($0.5 + 0.966\,i$) with lower E_{zp} than the triangular lattice, although its static minimiser is triangular. These are *detection-power* controls: they show that the search can find a stable lattice with lower E_{zp} when one exists. They are not false-positive controls; that role is played by the survivors (`gauss:2`, `gauss:4`, `gauss:0.25`, `riesz:6`, for which no stable lower lattice was found, `gauss:2` in particular at the competitor shapes of `gauss:1`) and by the frozen-phonon checks of (a).

4.1 Result 1: the counterexample

For `gauss:1` the rhombic lattice $\tau = 0.5 + 0.966025\,i$ has $\Delta E_{\mathrm{zp}} = -1.3074 \times 10^{-2}$, i.e. $E_{\mathrm{zp}} = 1.4902783183$ against 1.5033523522 (Table 2). It is stable with a clear margin (0.03493 on the zone, 0.1047 at long wavelength, against 0.03222 and 0.2383 for the triangular lattice). The G1 grid code at $n = 768$ gives 1.490278316 and 1.503352350 (differences from the spectral values 2.6×10^{-9}, 2.5×10^{-9}); the margin is 10052 times the $n = 96$ grid error at the triangular point (1.3×10^{-6}). At paper stage the compiled binary reproduced 1.490278318280 and 1.503352352207 ($\Delta E_{\mathrm{zp}} = -1.3074 \times 10^{-2}$). Across all 625 shapes sampled for this potential at $n = 96$ (Figure 2), 65 are dynamically stable; all 64 of the 64 stable non-triangular shapes have E_{zp} below the triangular value (weakest margin -1.3×10^{-4}), so around ρ the triangular lattice is a local *maximum* within the stable region.

Correction to a G1 figure. G1 reported its largest margin at $\tau = 0.295 + 0.9555\,i$ ($\Delta E_{\mathrm{zp}} = -5.123 \times 10^{-2}$). That value of E_{zp} is confirmed (Duffy, $n = 768$ and Richardson agree to about 10^{-10}), but the lattice is stable only marginally: its exact long-wave eigenvalue is 1.317×10^{-4}, 0.06 % of the triangular value, and the Duffy rule converges slowly there (change 3.3×10^{-9} from $N = 96$ to 128). It is *not* the headline.

4.2 Result 2: which potentials

QU-1 is refuted for 8 of 12 (`dipolar:0.1`, `dipolar:0.5`, `gauss:0.5`, `gauss:1`, `riesz:3`, `riesz:4`, `yukawa:1`, `yukawa:2`) by a stable lattice with lower E_{zp}; it is not refuted for `gauss:0.25`, `gauss:2`, `gauss:4`, `riesz:6` on the 567-point scan and the dense paths, where the triangular lattice is a strict local minimum of E_{zp} (an evidence statement from a sample and two one-dimensional paths, not an exhaustive search). Of the eight refutations, 7 also have a stable competitor below the triangular lattice at $0.5 + 0.966\,i$; the exception is `riesz:4`, whose refutation rests on one competitor ($0.3 + 0.954\,i$) with a small long-wave margin (0.08 % of the triangular value, Table 3) and whose triangular point is a local minimum of E_{zp} ($c > 0$); it is the weakest of the eight. The competitors at $x = 0.3$ for the dipolar, Riesz and Yukawa potentials have long-wave margins of 0.1 % to 1.5 % of the triangular value (Table 4); their finite-$\mathbf{k}$ stability rests on the G1 grids (n up to 384), since

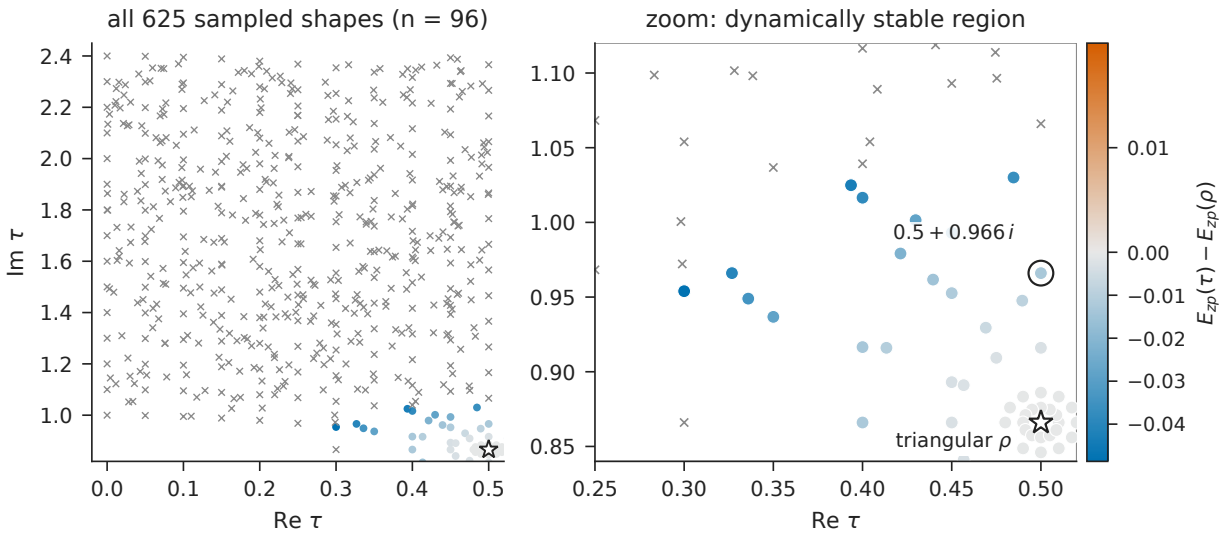


Figure 2: $E_{\mathrm{zp}}(\tau) - E_{\mathrm{zp}}(\rho)$ for `gauss:1` at $n = 96$ [X]. Left: all sampled shapes (grey crosses are dynamically unstable). Right: zoom on the dynamically stable region; the star is ρ and the ring is the headline $0.5 + 0.966\,i$. Blue is lower than triangular, orange higher (none). Generated by `make_figures.py` from the G1 scan files in `data/`.

the Poisson method is Gaussian-only. The refutation of "every completely monotone g" needs one potential, and `gauss:1` is verified by independent methods. Stable lattices form a small neighbourhood of ρ (within about 0.23 in τ at $n = 96$). Several lowest $n = 96$ competitors (`gauss:0.5` at $0.3 + 0.954\,i$, `yukawa:1` at $0.295 + 0.9555\,i$, `riesz:3` at $0.5 + 1.036\,i$) turned unstable on finer grids and were dropped; the first two are long-wave unstable (-3.81×10^{-4}, -9.50×10^{-4}), consistent with dropping them.

4.3 Result 3: curvature at the triangular point, and the Gaussian family

For $V = e^{-\pi a r^2}$ the curvature $c(a) = \frac{1}{4}\Delta E_{\mathrm{zp}}(\rho)$ (Step 6) is numerically negative, that is, the triangular lattice is a strict local *maximum* of E_{zp}, on $a \in (0.37668, 1.58148)$ within the range $0.15 \le a \le 4$. The sign changes were located by a scan of 23 values of a (0.1 to 8) and bisected (roots robust to the quadrature node count and the stencil step); the scan does not exclude further sign changes between its points, and at the roots themselves $c = 0$ and the second-order test is inconclusive. It contains the refuted `gauss:0.5` ($c = -0.461$) and `gauss:1` ($c = -0.962$); the survivors lie outside ($+0.188$, $+0.495$, $+0.313$ at $a = \frac{1}{4}, 2, 4$). The real-space Rust code at $n = 768$ confirms both sign changes independently: $E_{\mathrm{zp}}(\rho \pm 0.02) - E_{\mathrm{zp}}(\rho) = +4.220 \times 10^{-5}$ at $a = 0.35$, -4.279×10^{-5} at 0.40, -7.064×10^{-5} at 1.50 and $+5.404 \times 10^{-5}$ at 1.65. At ρ, for the four Gaussians with $a \ge 0.5$ of the mechanism study ($a = 0.5, 1, 2, 4$), $\partial_{xx} E_{\mathrm{zp}} = \partial_{yy} E_{\mathrm{zp}}$ to 1.5×10^{-6} relative, $|\partial_{xy} E_{\mathrm{zp}}| \le 2.0 \times 10^{-8}$ and $|\nabla E_{\mathrm{zp}}| \le 1.5 \times 10^{-6}$, as Step 6 predicts. At $a = 0.25$, near the soft-shear region, the relative difference is 4.3×10^{-4} and $|\nabla E_{\mathrm{zp}}| \approx 1.2 \times 10^{-5}$ (the finite-difference stencil is less accurate there). Table 5 also lists $\Delta\langle \mathrm{tr}\, D\rangle(\rho)$ from the exact lattice sum of Step 4; its sign changes at $a = 1.09361$, consistent with Luo–Wei ($a \le 1$) and with the assertion, quoted by Luo and Wei from Bétermin [4], that the hexagonal optimality of $S(\alpha; \cdot)$ cannot hold for all $\alpha > 0$ (here it fails locally for $\alpha < 0.914$). **The sum rule alone does not fix the sign of** c: they differ on about $(0.12, 0.38)$ and $(1.09, 1.58)$ (Figure 3).

4.4 Result 4: where the energy is lost

Finite-difference decomposition (Step 8) at $a = 1$ gives $\Delta E_{\mathrm{zp}}(\rho) = -3.848 = \mathrm{Lin}\,(-0.799) + \mathrm{Conc}\,(-3.048)$, with 63 % from the outer (zone-boundary) shell of the Duffy map; Lin + Conc matches the stencil Laplacian to 6.5×10^{-5}. At $a = 0.25$ and 2 a large positive Lin beats Conc (Table 6). For the headline competitor the loss is a *softening of the lower branch near the zone boundary* (-1.497×10^{-2} of -1.751×10^{-2}; long-wave shell

Table 3: Verdicts for the 12 completely monotone potentials [X, ZP-X-0003]. "Robust min" is the lowest $\Delta E_{\rm zp}$ at $n = 384$ among competitors stable on $n = 96, 192, 384$ (candidate points and the dense-scan minima); LW is the exact long-wave eigenvalue of that competitor as a percentage of the triangular value (– if not computed); the fifth column is $\Delta E_{\rm zp}$ at $0.5 + 0.966025\,i$ ($n = 384$, rc 400). †: for `gauss:1` the lowest robust competitor is the marginally stable point $0.295 + 0.9555\,i$ (long-wave margin 0.06 %), which the text demotes (Result 1); the refutation of `gauss:1` uses $0.5 + 0.966025\,i$ (fifth column; Tables 2, 4). Last column: range over 12 directions at $r = 0.01$ of $c = [E_{\rm zp}(\tau) - E_{\rm zp}(\rho)]y_\rho^2/|\tau - \rho|^2$ (extrapolated in n), and the resulting nature of the triangular point.

potential	robust min $\Delta E_{\rm zp}$	at τ	LW %	at $0.5{+}0.966i$	verdict	c at ρ
`dipolar:0.1`	-3.344×10^{-2}	$(0.300, 0.9539)$	1.02	-5.98×10^{-3}	**REFUTED**	$-0.33 \ldots -0.32$ (max)
`dipolar:0.5`	-4.068×10^{-2}	$(0.300, 0.9539)$	1.31	-9.01×10^{-3}	**REFUTED**	$-0.61 \ldots -0.58$ (max)
`gauss:0.25`	6.230×10^{-6}	$(0.500, 0.8710)$	94.64	2.08×10^{-3}	survives	$+0.19 \ldots +0.19$ (min)
`gauss:0.5`	-1.459×10^{-2}	$(0.500, 1.0160)$	1.71	-6.31×10^{-3}	**REFUTED**	$-0.47 \ldots -0.45$ (max)
`gauss:1`	-5.123×10^{-2}	$(0.295, 0.9555)^\dagger$	0.06	-1.31×10^{-2}	**REFUTED**	$-0.98 \ldots -0.95$ (max)
`gauss:2`	1.629×10^{-5}	$(0.500, 0.8710)$	96.56	4.53×10^{-3}	survives	$+0.48 \ldots +0.51$ (min)
`gauss:4`	1.039×10^{-5}	$(0.500, 0.8710)$	94.64	3.61×10^{-3}	survives	$+0.31 \ldots +0.32$ (min)
`riesz:3`	-3.274×10^{-2}	$(0.300, 0.9539)$	1.00	-5.71×10^{-3}	**REFUTED**	$-0.31 \ldots -0.29$ (max)
`riesz:4`	-1.238×10^{-2}	$(0.300, 0.9539)$	0.08	1.62×10^{-3}	**REFUTED**	$+0.34 \ldots +0.36$ (min)
`riesz:6`	1.134×10^{-4}	$(0.500, 0.8710)$	96.79	3.80×10^{-2}	survives	$+3.38 \ldots +3.45$ (min)
`yukawa:1`	-2.324×10^{-2}	$(0.300, 0.9539)$	1.54	-5.51×10^{-3}	**REFUTED**	$-0.40 \ldots -0.38$ (max)
`yukawa:2`	-1.972×10^{-2}	$(0.300, 0.9539)$	1.37	-4.60×10^{-3}	**REFUTED**	$-0.32 \ldots -0.31$ (max)

Table 4: Exact long-wave limit $\min_t \lambda_{\min} C(t)$ (no $\mathbf{k}$-grid; $r_c = 200$ and 400 agree to $\leq 9.0 \times 10^{-7}$) for one competitor sufficient for each refutation, the triangular lattice and the square lattice [X, ZP-X-0003]. The competitor is the robust minimum of Table 3 for every potential except `gauss:1`, where that minimum is the demoted marginal point $0.295 + 0.9555\,i$ and the table uses $0.5 + 0.966\,i$ instead.

potential	competitor τ	competitor	triangular	square
`dipolar:0.1`	$0.3 + 0.954\,i$	1.665×10^{-2}	1.6347	−0.9727
`dipolar:0.5`	$0.3 + 0.954\,i$	1.414×10^{-2}	1.0763	−0.6422
`gauss:0.5`	$0.5 + 1.016\,i$	1.379×10^{-3}	0.0808	−0.0449
`gauss:1`	$0.5 + 0.966\,i$	1.047×10^{-1}	0.2383	−0.1475
`riesz:3`	$0.3 + 0.954\,i$	1.668×10^{-2}	1.6674	−0.9919
`riesz:4`	$0.3 + 0.954\,i$	2.258×10^{-3}	2.8917	−1.7134
`yukawa:1`	$0.3 + 0.954\,i$	3.454×10^{-3}	0.2250	−0.1343
`yukawa:2`	$0.3 + 0.954\,i$	2.401×10^{-3}	0.1753	−0.1047

-1.33×10^{-5}), partly offset by the upper branch (4.436×10^{-3}) (Figure 4). The mean-square frequency falls by $|-2.033 \times 10^{-2}|$, which moves the Einstein bound by only -3.131×10^{-3} (24 % of the gain); the other 76 % (-9.943×10^{-3}) is the Jensen gap, a wider spread of the spectrum. The negative control `gauss:2` at the same shape has $\langle \mathrm{tr}\, D \rangle$ rising by 5.393×10^{-2}, the upper branch gaining 2.440×10^{-2}, and a net 4.533×10^{-3}. These are Tier X numbers with a Tier C reading (zone-boundary softening of the shear branch; ZP-X-0007, ZP-C-0009): the mechanism was not tested separately.

4.5 Result 5: the full functional $E_{\rm static} + \Lambda E_{\rm zp}$

With $V \to \epsilon V$, $E_{\rm static}$ scales as ϵ and $E_{\rm zp}$ as $\sqrt{\epsilon}$, so the physical family is $E_{\rm static} + \Lambda E_{\rm zp}$ with $\Lambda \propto \epsilon^{-1/2}$ (ZP-X-0010). Over the stable sample the triangular lattice stops being the minimiser at some $\Lambda \in [0.1, 10.0]$ for every refuted potential (for `riesz:4` only at $\Lambda = 10$), and remains the minimiser for the survivors at all five $\Lambda \in \{0.1, 0.3, 1, 3, 10\}$. The harmonic approximation fails for large Λ (strong quantum motion, melting), so this is a statement about the functional only. Moreover the competitors are *not stress-free*: the triangular lattice has the lower static energy, so each competitor is held in shape by a constraint; a statement about $E_{\rm total}$ at its own equilibrium shape is a different statement and is untested.

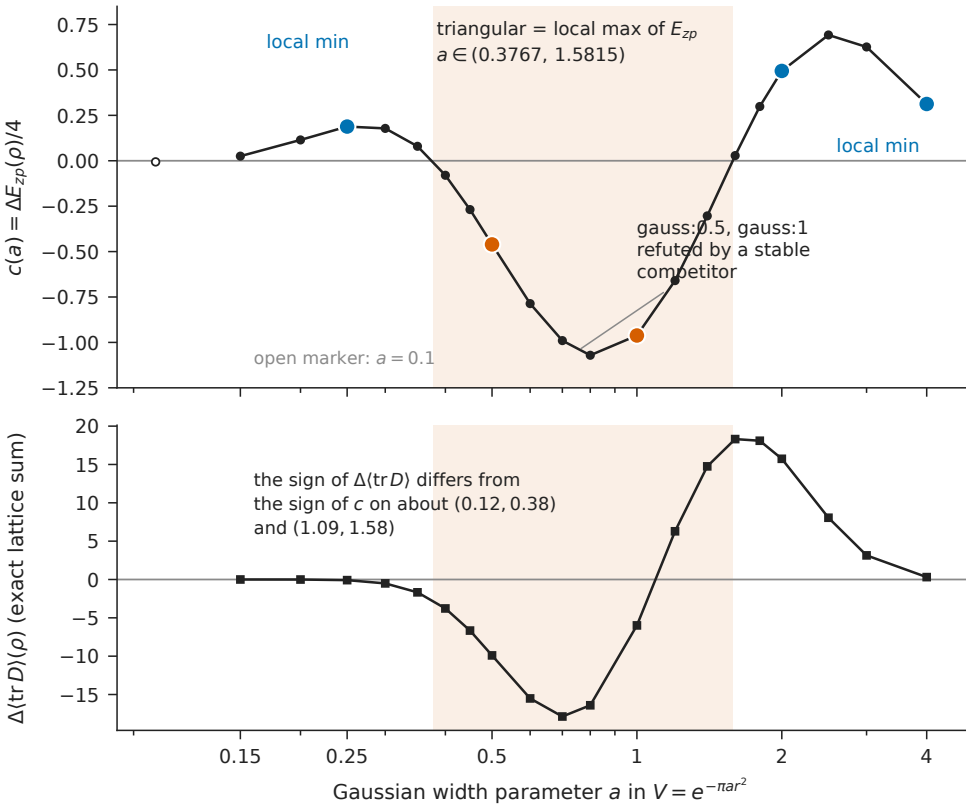


Figure 3: Top: $c(a) = \Delta E_{\mathrm{zp}}(\rho)/4$ for $V = e^{-\pi a r^2}$ [X]; shaded: $c < 0$, the triangular lattice is a local maximum of E_{zp}. Filled markers are the five Gaussians of the G1 scan (orange: refuted). Bottom: $\Delta\langle \mathrm{tr}\, D\rangle(\rho)$, the exact lattice sum of Step 4 [X, with Tier A formula]. Generated by `make_figures.py` from `data/mechanism.json`.

4.6 The rusty-SUNDIALS improvement

The lane added `phonon.rs` (the lattice-phonon module: real-space $D(\mathbf{k})$, folded-grid E_{zp}, stability, Epstein/Ewald statics; 12 unit tests according to the lane, incl. a regression of the headline and a negative control) and, in this stage's commit `6ad3cf7`, `phonon_spectral.rs` (280 lines): Gauss–Legendre nodes, `GaussPoisson` (reciprocal-space Gaussian dynamical matrix with the exact sum rule), `ezp_gauss_duffy` (the spectral zone rule) and `curvature_at_rho_gauss`; plus the `lattice_phonon` subcommands `duffy` and `curv`. The spectral rule replaces an n^{-3} grid error (about 10^{-6} at $n = 96$) by about 4.4×10^{-15} at $N = 128$ (Figure 1), which makes the sign of $c(a)$ resolvable to its roots and turned a numerical refutation into one whose margin is about 10^{12} times the quadrature error (the remaining uncertainty is modelling: the harmonic approximation). Five new unit tests cover Gauss–Legendre exactness, Poisson against real-space D at three shapes to 10^{-12}, Duffy against Richardson grids, a regression of the headline with the square lattice as negative control, and the sign of c at $a = 0.25, 1, 2$. At paper stage (same pipeline, `data/rerun_*`): `cargo test -p qf-yukawa-crystal`: 59 passed, 0 failed, 1 ignored; `cargo clippy -workspace -features experimental - -D warnings`: no warnings or errors; `cargo fmt -check`: clean; `lattice_phonon curv 1 48 0.002` gives $c = -0.961984$. After the audit (hardening point N7) commit `609338e` added `ezp_gauss_duffy_stable` and `DuffyResult::is_stable`, so that the clipped E_{zp} of an unstable shape (which the unchecked rule returns, favouring the unstable lattice) cannot be used by accident, and doc comments saying so; no existing function changed behaviour, one test was added (stable shape returns the same energy, the square lattice returns `None`); `cargo test -p qf-yukawa-crystal`: 60 passed, 0 failed, 1 ignored; fmt and clippy clean (`data/revise_cargo_*.txt`). The reported numbers were all produced before this commit. No LeanFlow change was made in this lane.

Table 5: $c(a)$, the curvature of $\langle \mathrm{tr}\, D \rangle$ from the exact lattice sum, and the smallest $\lambda/|\mathbf{k}|^2$ of the triangular lattice, for $V = e^{-\pi a r^2}$ [X, ZP-X-0005]. † the triangular lattice is marginal (shear eigenvalue $\leq 1.5 \times 10^{-9}$ for $a \leq 0.123$, 3.3×10^{-12} at 0.1; harmonic picture doubtful; c changes sign again near $a \approx 0.123$, uncertain by about 10^{-3} since the quadrature rules shift c by 10^{-5} there). ‡ precision-flagged: the Poisson form loses relative accuracy when E_{zp} is tiny; at $a = 8$ the triangular lattice failed the stability test.

a	c	$\Delta\langle \mathrm{tr}\, D \rangle$	$\min \lambda/k^2$	a	c	$\Delta\langle \mathrm{tr}\, D \rangle$	$\min \lambda/k^2$
$0.1^\dagger$	-0.0062	-5.6×10^{-10}	3.3×10^{-12}	1.2	-0.6598	$+6.3 \times 10^{0}$	2.7×10^{-2}
0.15	$+0.0254$	-2.8×10^{-5}	1.7×10^{-7}	1.4	-0.3035	$+1.5 \times 10^{1}$	2.0×10^{-2}
0.2	$+0.1147$	-4.7×10^{-3}	1.2×10^{-5}	1.6	$+0.0285$	$+1.8 \times 10^{1}$	1.3×10^{-2}
0.25	$+0.1882$	-8.4×10^{-2}	1.4×10^{-4}	1.8	$+0.2987$	$+1.8 \times 10^{1}$	8.7×10^{-3}
0.3	$+0.1781$	-5.1×10^{-1}	6.9×10^{-4}	2	$+0.4945$	$+1.6 \times 10^{1}$	5.5×10^{-3}
0.35	$+0.0796$	-1.7×10^{0}	2.0×10^{-3}	2.5	$+0.6925$	$+8.1 \times 10^{0}$	1.5×10^{-3}
0.4	-0.0799	-3.8×10^{0}	4.4×10^{-3}	3	$+0.6265$	$+3.1 \times 10^{0}$	3.7×10^{-4}
0.5	-0.4605	-9.9×10^{0}	1.2×10^{-2}	4	$+0.3125$	$+3.1 \times 10^{-1}$	1.8×10^{-5}
0.6	-0.7864	-1.6×10^{1}	2.0×10^{-2}	$6^\ddagger$	$+0.0345$	$+1.3 \times 10^{-3}$	3.1×10^{-8}
0.8	-1.0707	-1.6×10^{1}	3.1×10^{-2}	$8^\ddagger$	$+0.0025$	$+3.1 \times 10^{-6}$	4.0×10^{-11}
1	-0.9620	-6.0×10^{0}	3.2×10^{-2}				

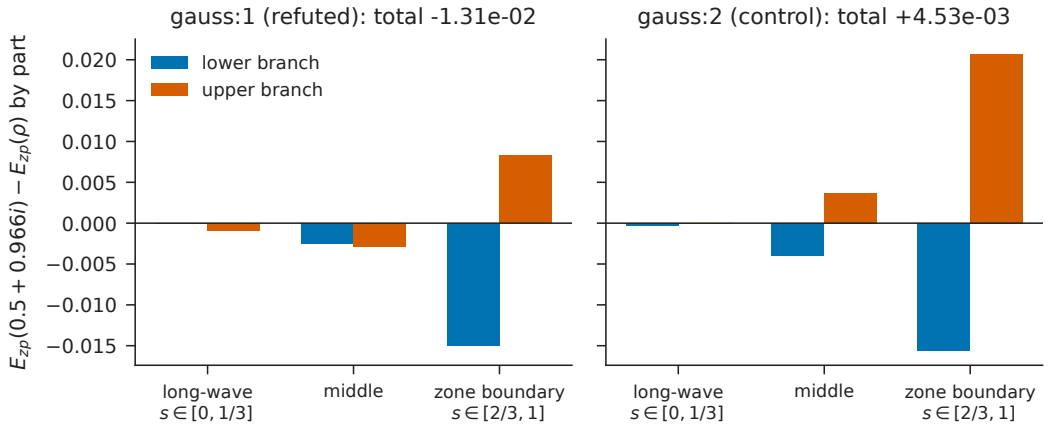


Figure 4: $E_{\mathrm{zp}}(0.5 + 0.966\, i) - E_{\mathrm{zp}}(\rho)$ split by branch and by zone shell (Duffy radial thirds) [X]; left the refuted `gauss:1`, right the control `gauss:2`. Generated by `make_figures.py` from `data/mechanism.json`.

5 A restricted conjecture, QU-1′

QU-1′ [X-supported conjecture, not proved]. For $g_a(t) = e^{-\pi a t}$ with $0.15 \leq a \leq 4$, the triangular lattice is a strict local minimiser of E_{zp} among unit-covolume planar Bravais lattices if $a < 0.37668$ or $a > 1.58148$, and a strict local maximiser (so not a minimiser) for $0.37668 < a < 1.58148$; the numerical c of Table 5 is negative there. The endpoints $a_1 = 0.37668$ and $a_2 = 1.58148$ are bisected numerical roots of c; at $c = 0$ the second-order test is inconclusive, so nothing is stated at the endpoints, and the roots carry the uncertainty of a floating-point scan. It is suggested by Table 5 and supported by the $n = 768$ cross-check; nothing proves it. Caveats kept outside the statement: at $a = 0.15$ the triangular lattice is barely stable ($\min \lambda/|\mathbf{k}|^2 = 1.7 \times 10^{-7}$); below $a \approx 0.123$ the sign of c changes again at a marginal lattice; $a = 6, 8$ are precision-flagged. The global version (triangular is the E_{zp} minimiser among dynamically stable lattices for $a > 1.58148$) is supported only by the G1 sample at $a = 2, 4$ and is open. A proof would need rigorous bounds on the Lin and Conc integrals, for instance by interval arithmetic over the Duffy rule; $\Delta E_{\mathrm{zp}}(\rho)$ has no theta-function closed form because E_{zp} is not a pair sum. This was not attempted.

Table 6: Decomposition of $\Delta E_{\mathrm{zp}}(\rho)$ into linear response and concavity (diagonal and branch-mixing parts), and the curvature of the Einstein bound [X, ZP-X-0007].

a	ΔE_{zp}	Lin	Conc	Conc (diag)	Conc (cross)	$\Delta E_{\mathrm{Einstein}}$
0.25	+0.753	+9.683	−8.930	−0.506	−8.425	−0.017
0.5	−1.842	+0.028	−1.870	−0.733	−1.137	−1.441
1	−3.848	−0.799	−3.048	−1.189	−1.859	−0.921
2	+1.978	+6.400	−4.422	−1.644	−2.777	+6.817
4	+1.250	+2.294	−1.044	−0.396	−0.648	+2.432

6 Claim ledger

`ledger/build_ledger.py` files 37 claims: 8 at Tier A, 3 at L, 13 at C and 13 at X. Each has a content-addressed evidence blob in `ledger/evidence/`: verbatim axiom lines and source hashes (A), the quotation with the Elenchus citation-gate result (L), the argument text (C), and result files with `claim_licence: false` (X). The tier discipline of `tools/ledger.py` (a claim may not sit above anything it depends on) forces everything resting on a floating-point result to X: the sum-rule consequences that combine Luo–Wei with a Poisson identity are filed at C, not L; the mechanism reading, the roots a_1, a_2 and QU-1′ are filed at X, not C. The gate (`tools/ledger.py ledger.json --evidence-dir evidence`, verbatim output in `ledger/gate_output.txt`) exits with code 1. It reports **0 blocking findings** and **8 flags** (LEDGER_UNAUDITED_TIER_A, one per Tier A row: no person has certified that the Lean statements mean what their glosses say). The flags remain open. The Elenchus citation gate (`tools/citation_check.py`) *accepted* 3 of 3 quotations (tier ceiling L); the quotations were copied programmatically from `pdftotext` output of the arXiv PDFs, so the gate tests fidelity to the fetched text, while "statement match" is the agent's own judgement. Montgomery's theorem is cited through Deng–Luo [2]; the original paper was not opened.

7 Limitations

- **Everything about E_{zp} is Tier X or C.** The refutation is floating-point: robust (four implementations, stable on grids to $n = 384$ and by the exact long-wave limit, frozen-phonon agreement 2.14×10^{-7}), but not a Lean statement and not an interval-arithmetic enclosure. A certified enclosure of E_{zp} at the two shapes would upgrade it and is the natural next step; it was not done.
- **Harmonic, Bravais, two-dimensional only.** Nothing is said about non-Bravais structures (the harder QU-2), anharmonic corrections, quantum melting or the true quantum ground state; these matter most where the shear eigenvalue is tiny (small a, marginal points).
- **Competitors are not stress-free.** See Result 5. QU-1 as stated compares a functional over imposed shapes and is refuted regardless; E_{total} at equilibrium shape is untested.
- **Sample, not exhaustive search.** The survivors rest on 567 shapes plus two dense paths; the refutations rest on competitors sampled near ρ. We do not claim any listed point is the global minimiser of E_{zp}.
- **Stability margins.** $n = 96$ stability is optimistic near the edge. The $x = 0.3$ competitors have small long-wave margins; `riesz:4` rests on a single such point. Finite-**k** stability of the non-Gaussian competitors rests on G1's grids.
- **Excluded potentials.** Coulomb and Keldysh (neutralising background, plasmon $k \to 0$) are not covered. The harmonic zero-point energy of unit-amplitude potentials with $\hbar = m = 1$ only rescales; the sign of ΔE_{zp} does not depend on units.
- **Post-hoc analysis.** The curvature scan, the mechanism and QU-1′ were run after G1 refuted the conjecture and were not preregistered. The mechanism (zone-boundary softening) is interpretation.
- **Novelty.** One alphaXiv search found no paper on lattice-shape optimality of the harmonic zero-point energy: this is absence by one search, not a claim of novelty.
- **Verification status.** No human audit of this preprint, of the Lean module, or of the Rust code. One separate AI audit exists, with the scope of Section 8: several results (the dipolar, Riesz and Yukawa

refutations, the long-wave elastic tensor, the Lin/Conc split, Result 5, $c(a)$ beyond its sign at $a = 1, 2$, and the 691 lines of `phonon.rs`) were checked only through the producer's data and tests. The paper-stage reruns are by the same pipeline. The seed detector could not run. Nothing here rests on openai/math (AI-generated, not peer reviewed), except that its toolchain compiles the Lean file.

- **Literature.** Montgomery's 1988 paper and Bétermin's 2016 paper were not opened; both are cited second-hand through arXiv texts of Luo–Wei and Deng–Luo. The Luo–Wei journal version (J. Math. Phys. 65 (2024) 071901, as listed in Deng–Luo, ref. [27]) was not opened; we cite the arXiv text.

8 Separate AI audit: scope and result

After the paper stage, a separate AI agent (Claude; it wrote none of the paper, the Lean module, the Rust code or the analysis) audited the preprint; the report is `AUDIT.md`, its verdict is "accept with corrections" with no blocking finding, and the corrections were applied (`REVISION.md`). It is a producer $\neq$ verifier check at the agent level only; **no human audit exists**, and the eight open Tier A ledger flags stay open because an AI adequacy review is not a person's certification. *Recomputed with the auditor's own code:* the Lean module (recompiled with the pinned stack, 19 printed declarations plus the helper `rho_ne_zero` within the allowlist; the auditor's own planted `sorry` and planted axiom were caught by `#print axioms`; the stated derivatives are Mathlib's `deriv`; the hypothesis of the inversion lemma is satisfiable); E_{zp} of `gauss:1` at ρ and at $0.5 + 0.966025\,i$ by an independent real-space code sharing no code with the producer (agreement to about 10^{-12}, invariance under $\tau \mapsto -1/\tau$), stability signs of the triangular, rhombic and square lattices, and the sign and size of c at $a = 1, 2$; the `gauss:0.5` refutation at $(0.5, 1.016)$; the map recount of `gauss:1`; all generated macros and tables by regenerating them from `data/`; the three citations against the arXiv PDFs (the citation gate accepted all three); the crate tests, `cargo fmt` and clippy. *Checked only through the producer's data or tests, not recomputed:* the dipolar, Riesz and Yukawa refutations, the exact long-wave elastic tensor, the Lin/Conc decomposition, Result 5, $c(a)$ beyond its sign at $a = 1, 2$, and `phonon.rs`. The audit's reproduction of the ledger gate gave 0 blocking findings and 8 flags, as in Section 6.

9 Reproducibility

AutoevolveAI: branch `worktree-openai-math-discovery`, HEAD below; the new files (this paper, the Lean module, the G2 result files) were uncommitted when the PDF was built and are committed by the release procedure, not by this revision; the cited hashes of the Lean files below identify their content. Lean sha256 values are in Section 3.

- AutoevolveAI HEAD: `9d245742c1655852103abb28f585e21af54851e6`

rusty-SUNDIALS branch `physics/p2-zero-point-optimality-2026-10-25`, local, not pushed:

- `a961a714dd040ad1e1a542db1efc54e0401cd55b` (lattice-phonon module and falsification gate);
- `5f2f73b8c636c4ae45a43ad67c99621acc609660` (stability-edge verification, dense scans);
- `6ad3cf7e5e171479d1481e5fd17e7d432aae21d1` (spectral zone rule, curvature, G2 checks);
- `609338e6485fc96002e4c60c0c3e34f9130c0108` (post-audit hardening: `ezp_gauss_duffy_stable`, no change to any reported number).

```
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPoint/Basic.lean \
  --olean Programmes.ZeroPoint.Basic --json-out papers/prog_zero_point_optimality/data/rerun_basic.json
taskset -c 4-7 python3 scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/ZeroPoint/Controls.lean --json
     -out .../rerun_controls.json
CARGO_TARGET_DIR=... cargo test --release -j 2 --offline -p qf-yukawa-crystal -- --test-threads=2
CARGO_TARGET_DIR=... cargo clippy -j 2 --offline --workspace --features experimental -- -D warnings
lattice_phonon duffy 1 0.5 0.966025403784 64 ; lattice_phonon curv 1 48 0.002
python -I phonon_poisson_check.py | phonon_elastic_check.py | phonon_mechanism.py | phonon_smalla_check.py |
     phonon_sumrule_identity.py   # crates/qf-yukawa-crystal/scripts
python -I papers/prog_zero_point_optimality/human_check.py ; make_figures.py ; ledger/build_ledger.py ; ledger/run_gate.py ;
     gen_inputs.py ; pdflatex main.tex (twice)
```

All numbers in this paper are generated into `gen_numbers.tex` and `gen/` from the files in `data/`, which are copies of the lane's results in `results/programmes/2026-10-25/P2/` (`GATE.md`, `PROOF.md`).

References

[1] H. L. Montgomery, *Minimal theta functions*, Glasgow Math. J. 30 (1988) 75–85. Cited second-hand through [2], p. 3, Theorem A; the original was not opened.

[2] K. Deng, S. Luo, *Minimizing lattice energy and hexagonal crystallization*, arXiv:2411.17199v1 (2024). Accessed 2026-10-25; Theorem A quoted and gate-checked.

[3] S. Luo, J. Wei, *On lattice hexagonal crystallization for non-monotone potentials*, arXiv:2302.05042v1 (2023); journal version J. Math. Phys. 65 (2024) 071901 (listed as ref. [27] of [2]; not opened). Accessed 2026-10-25; Corollary 1.1 (p. 3) and Lemma 3.5 (p. 12) quoted and gate-checked. SHA-256 of the fetched texts: `ledger/sources/MANIFEST.md`.

[4] L. Bétermin, *Two-dimensional theta functions and crystallization among Bravais lattices*, SIAM J. Math. Anal. 48(5) (2016) 3236–269 (pages exactly as printed in [3], ref. [7]); cited only as described by Luo–Wei (their pp. 2–3, where Subsection 4.3 of Bétermin, p. 3249, is referred to); not opened.

[5] X. Callens, five autoresearch preprints on triangular-lattice corollaries and consistency studies: doi:10.5281/zenodo.23646103 (dipolar excitons), 23646105 (Yukawa), 23646107 (Rytova–Keldysh), 23646111 ($\sigma = 3$ dyadic shell model), 23646117 (Bogoliubov structure factor) (2026).

[6] X. Callens, *The Triangular Lattice and Universal Optimality* (textbook), doi:10.5281/zenodo.23295142 (2026).

This preprint applies the approach of [5, 6] (kernel-checked consequences, Elenchus tiers, rusty-SUNDIALS consistency studies) to the zero-point functional and reports a negative result.